IC improvement round 3

IC improvement round 3: results (2026-10-15)

Two follow-ups to `../ic improve 20261015/README.md`, run under `RULES.md`.

- Each candidate is measured against the incumbent component it replaces.
- Comparisons are paired, with training and held-out splits.
- Every logarithm or root set is verified.

#	Candidate vs incumbent	Verdict	Key numbers	Landed
01	Genus-3 IC relation phase vs round-2 item 06 `sched:lpsgebw2`. **R1:** cheaper per-trial primitives, with identical relations. **R2:** R1 plus a re-tuned κ.	R1 TRADES; R2 ADVANCES (constant)		example only
02	Koblitz Gröbner linear-row harvest: root-rank gate (`gpv` < `0.9607` or root rank ≥ 331) vs round-2 item 05's gpv gate	**ADVANCES** (constant), no regression		**Default** in `koblitz_groebner.rs`. `KIC_LINEAR_HARVEST=gpv` restores the previous gate; `=0` turns harvesting off.

What the genus-3 diagnosis changed

The relation phase is not what keeps the exponent above 4/3.

- **Trials grow below 4/3.** At fixed κ, the number of trials grows as q^1.258.
- **The log factor is small.** The per-trial log factor adds about 0.05 to the exponent.
- **The κ schedule drove the earlier figure.** The 1.43 seen in round 2 came from the falling κ schedule.
- **The schedule falls because of linear algebra.** κ drops because linear algebra grows faster than q^{4/3}. Roughly, the exponent at the optimal κ is 4/3 + (2/3)(per-trial log growth) + (1/3)(LA excess).
- **Next lever:** the linear-algebra exponent, currently about q^1.52–1.54 locally.

Caveats

- **Host:** a shared 4-vCPU container at isolation level L0.
- **Measured unit:** operation counts; wall-clock times are descriptive only.
- **Scale:** sizes are small. 2^23 projects to just over the 60-minute job cap.
- **Conductor:** the control plane was unreachable.